



Must-Know Cross-Cutting Concerns in API Development

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Thu, Apr 9, 2026 at 11:41 AM

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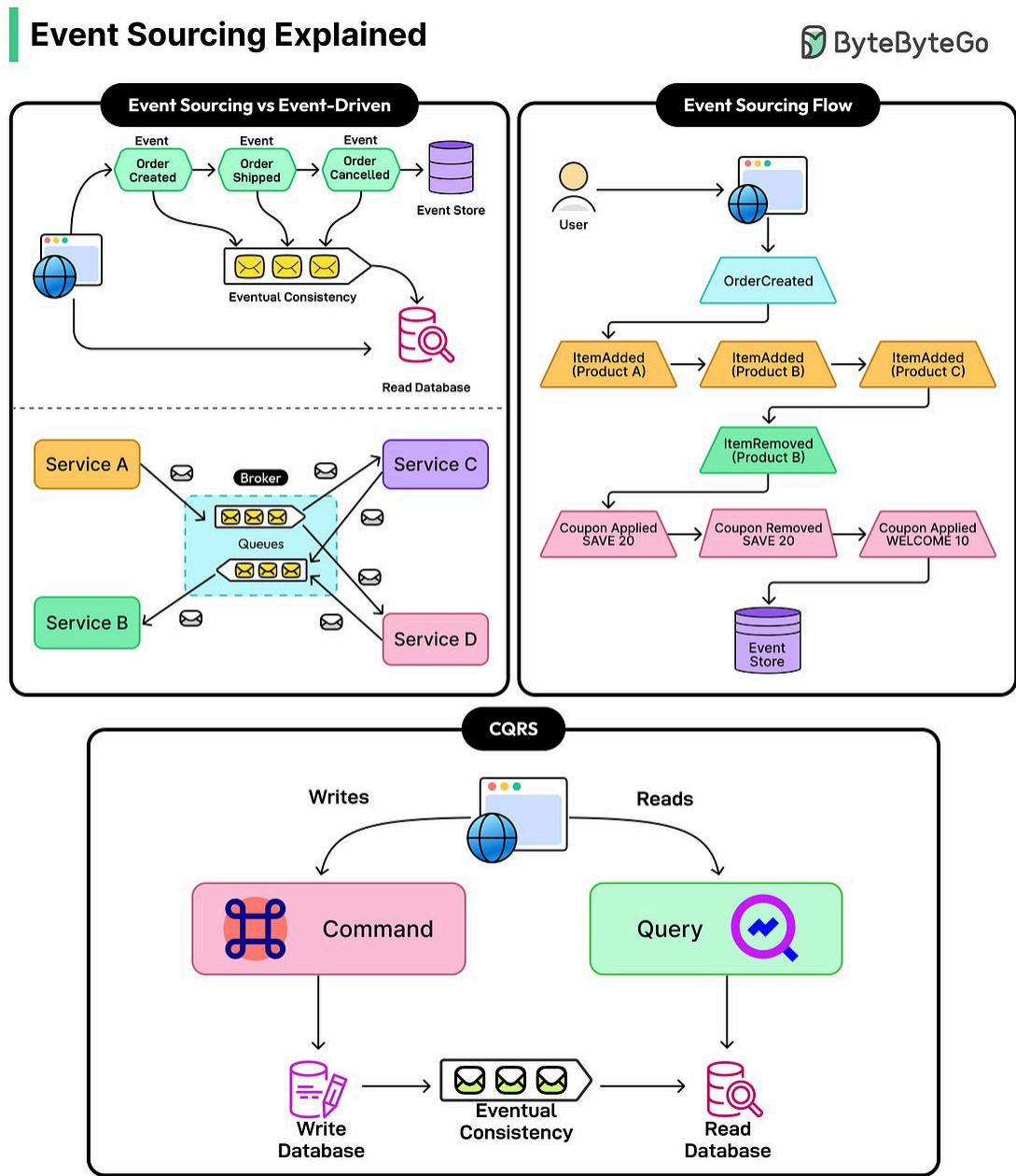
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What do authentication, logging, rate limiting, and input validation have in common?

The obvious answer is that they're all important parts of an API. But the real answer is deeper is that none of them belong to any single endpoint or show up in usual product requirements. For all purposes, they are invisible to users when they work and catastrophic when they're missing. And the hardest part about all of them is making sure they're applied uniformly across every single route an API exposes.

This family of problems has a name. They're called cross-cutting concerns, and they're the invisible layer that separates a collection of API endpoints from a production-ready system.

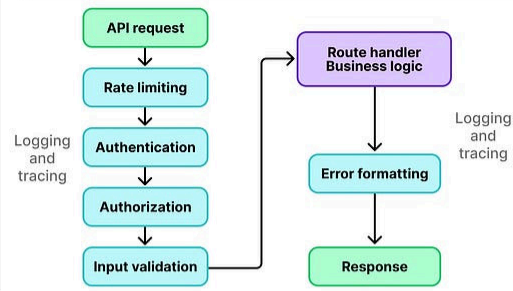
In this article, we will learn about these key concerns and their trade-offs in detail.

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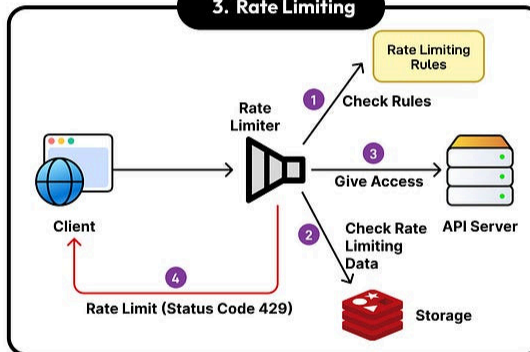
1. An Ideal Cross-Cutting Matrix

	GET /users	POST /orders	GET /products	DELETE /items
Authentication	✓	✓	✓	✓
Authorization	✓	✓	✓	✓
Input validation	✓	✓	✓	✓
Rate limiting	✓	✓	✓	✓
Caching	✓	✓	✓	✓
Logging	✓	✓	✓	✓
Error handling	✓	✓	✓	✓

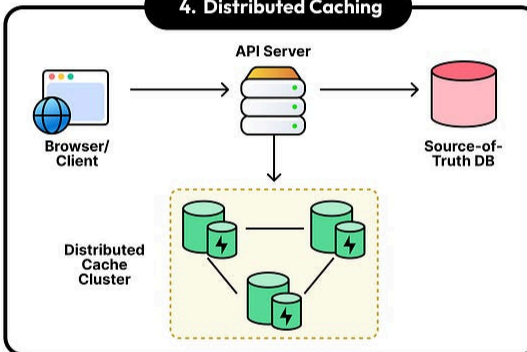
2. Production Request with Cross Cutting Concerns



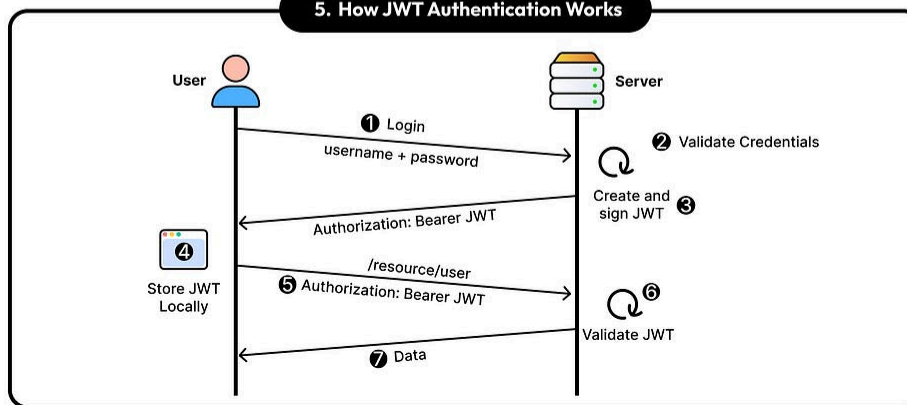
3. Rate Limiting



4. Distributed Caching



5. How JWT Authentication Works



What Makes a Concern “Cross-Cutting”...



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